

PRACTICAL COURSE WORKBOOK

Python for Finance & Quant Analysis

Build a clearer, auditable financial workflow

LEVEL	Intermediate
GUIDED TIME	5 hours across 12 lessons
PROJECT	a fictional financial model with assumptions and checks
SKILLS	Python, yfinance, NumPy, Portfolio, Quant, Critical evaluation
EDITORIAL STATUS	Structured draft v0.9

WHAT YOU WILL BE ABLE TO DO

- Use Python appropriately in a realistic, bounded task.
- Use yfinance appropriately in a realistic, bounded task.
- Use NumPy appropriately in a realistic, bounded task.
- Use Portfolio appropriately in a realistic, bounded task.

WHO THIS IS FOR

Learners studying financial technology and analytical methods.

BEFORE YOU BEGIN

- Basic percentages and spreadsheet skills

This open workbook is a structured draft undergoing course-specific editorial review. It does not provide certification.

COURSE MAP

From foundation to finished work

Python for Finance & Quant Analysis is a structured, practical course covering Python, yfinance, NumPy, Portfolio, Quant. It emphasizes explainable methods, checked work and a final outcome that can be reviewed.

MODULE	LESSONS AND FOCUS
01 Python: Financial foundations	1. Value with Python 2. Risk with yfinance 3. Markets with NumPy
02 yfinance: Tools and data	1. Records with Portfolio 2. Analysis with Quant 3. Controls with Python
03 NumPy: Applied decisions	1. Scenarios with yfinance 2. Evaluation with NumPy 3. Reporting with Portfolio
04 Portfolio: Risk and governance	1. Security with Quant 2. Regulation with Python 3. Review with yfinance

HOW TO STUDY EACH LESSON

- 1 Read the objective and identify the decision it supports.
- 2 Reproduce the worked example with the stated input.
- 3 Test one normal case and one boundary or failure case.
- 4 Complete the knowledge check without guessing.
- 5 Save a short evidence note: result, limitation and next action.

CAPSTONE PROJECT

Produce a reviewable Python for Finance & Quant Analysis project using Python, yfinance, NumPy.

DELIVERABLE	A working Python for Finance & Quant Analysis artefact plus an evidence-based self-review.
TOOLS	A suitable Python environment, A plain-text decision log, Test data or realistic sample material

Production plan

- 1 Define the intended user, outcome and constraints.
- 2 Create the smallest complete result using Python.
- 3 Test one normal case, one boundary case and one failure response.
- 4 Revise the work from the evidence and preserve before-and-after results.
- 5 Prepare a concise handover containing method, limitations and next step.

Definition of done

- The output matches the stated outcome.
- Inputs and decisions are reproducible.
- Boundary and failure evidence is included.
- Limitations and responsibility considerations are explicit.

RESPONSIBLE PRACTICE

This course is general education, not financial, investment, tax or legal advice.

Evidence log

INPUT	Original brief, data, requirements or test material.
DECISIONS	Chosen method and one reasonable alternative.
TESTS	Normal, boundary and failure results.
LIMITATION	What the result does not establish.
REVISION	One evidence-led improvement.

REVIEW AND SOURCES

Make the work credible and repeatable

Final self-review

- Can another learner repeat the method?
- Did I test a difficult case?
- Did I avoid unsupported claims?
- Is the next action proportionate to the remaining risk?

Authoritative references

National Payments System

Central Bank of Kenya - reviewed 2026-08-21
<https://www.centralbank.go.ke/national-payments-system/>

Financial education

OECD - reviewed 2026-08-21
<https://www.oecd.org/en/topics/financial-education.html>

SOURCE NOTE

Product versions, laws and professional standards can change. Recheck the linked primary source before applying version-sensitive information.

NEXT IMPROVEMENT

Choose one weakness found during review and improve it before extending the Python scope.